Software testing

Agenda

- Software testing definition
- Purpose of Software testing
- Objectives of software testing
- Testing strategies
- Levels of software testing

Software testing

- It is the process of identification and rectification of errors in the developed system.

or

- It is the process of executing a program with the intent of finding errors.

or

- It is the process of evaluating software quality and also improving its functionality.

Software testing purpose

The purpose of software testing is *software verification and validation*.

- **Verification** is the checking of the software conformance and consistency by evaluating the results against the predetermined requirements.
- **Validation** is checking the correctness of the software that is if the correct system has been developed.

Objectives of software testing

- The main objectives of software testing are; prevention, detection and improving quality.
  - **Prevention** is the avoidance or reduction of number of errors, to clarify system specifications and system performance;
  - **Detection** is the process of finding of errors, defects, deficiencies in the software; **improving quality** is achieved through an effective testing to minimize software errors.
    - **Error detection** is the process of finding errors in the software.

Testing strategies

i. Grey box testing

- It is also known as ***translucent testing***.
- *Grey Box testing* is a technique to test the software with limited knowledge of the internal workings of a system.

Testing strategies

ii. Black box testing

- This is also known as ***closed box testing.***
- This is the testing of general software functionality without looking into the internal code structure.
- Different categories of black box testing exist, namely; Equivalent partitioning, Boundary value analysis, Cause-effect graphing techniques, Comparison testing fuzz testing and Model-based testing

Testing strategies

White box testing:

- It is also referred to as ***clear box.***
- It is the checking of the software internal functionality that is its internal coding and infrastructure.
- White box types include; Basis path testing, Loop testing and Control structure testing.

Levels of software testing

i. Unit testing:

- This is the lowest level of software testing where the smallest lines of codes known as ***module*** or ***unit*** is tested.
- It normally focuses on the verification of the modules.

Levels of software testing

ii. Integration testing:

- This is a systematic technique for verifying the software structure and sequence of execution while performing tests on the software.
- Two or more modules are put together or assembled into larger structure as the test is done.
- Integration testing can be sub-divided as follows; Big-bang approach, Top-down approach, Bottom-up approach and Sandwich (Mixed integration) approach.

Levels of software testing contd

iii. System testing:

- System testing checks the overall functioning of the system by testing the quality of the entire developed system.
- It checks if the system conforms to its specified functional and non-functional requirements.

Levels of software testing contd

iv. Acceptance testing:

- This type of testing is performed when the system is handed over to the end user.
- It main aim is to assure the end user of the system that the software is working well.
- Acceptance testing can be done through ***alpha*** and ***Beta*** testing.

Levels of software testing contd

iv. Acceptance testing cont..d:

- **Alpha testing** is the testing done by the development team while
- **Beta testing** is normally done by the end user of the system after the system has been delivered.